

Section B

Answer **all** questions in the spaces provided.

2. Medical imaging techniques include ultrasound, X-rays, CAT scans and MRI scans.

(a) (i) Describe how the two strands of DNA are joined in a double helix. [2]

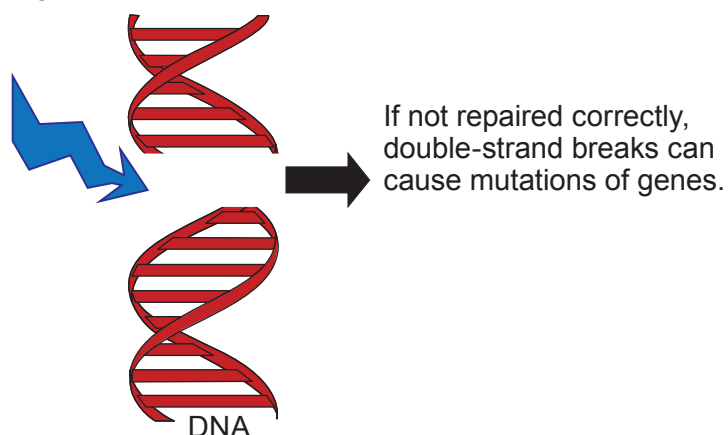
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(ii) Ionising radiation can have harmful effects on DNA within a human body. The diagram shows that it can cause a break in the double-strand.

Double-strand breaks produced
by ionising radiation, etc.



It is suggested that MRI scans are most likely to cause this kind of DNA damage and the other methods listed above are harmless. Explain whether you agree. [3]

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(b) Describe how an ultrasound image is formed. [2]

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